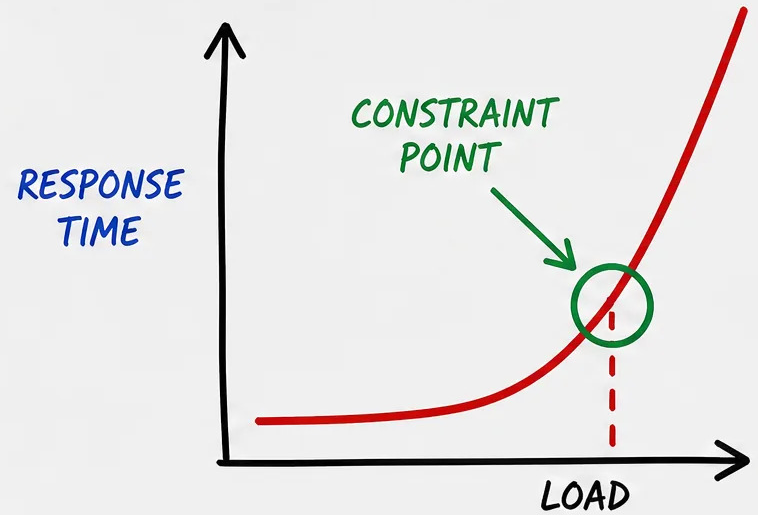
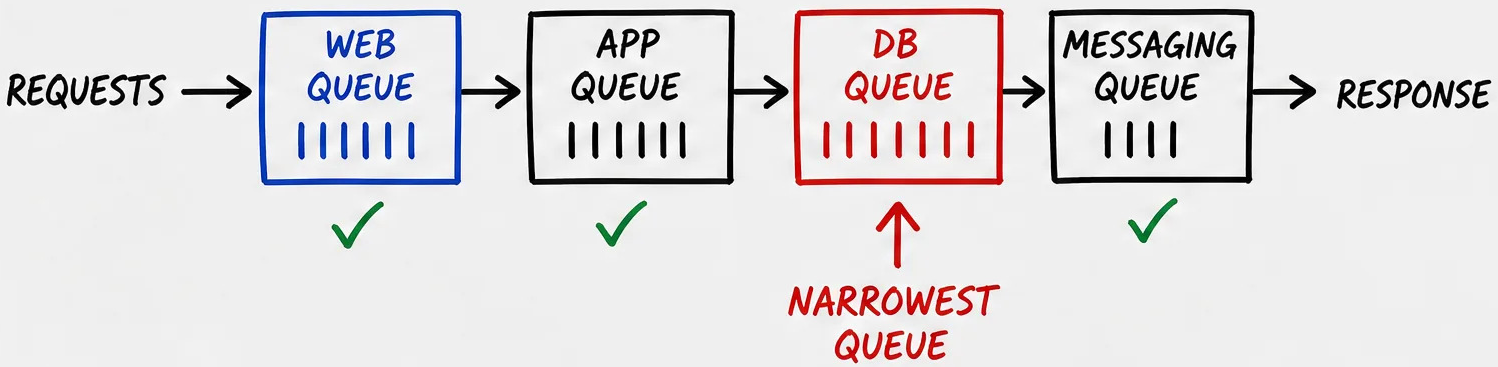
Module 15 — WebSphere Application Server Performance Tuning

ProTech
protechtraining.com

Find The Narrowest Queue.

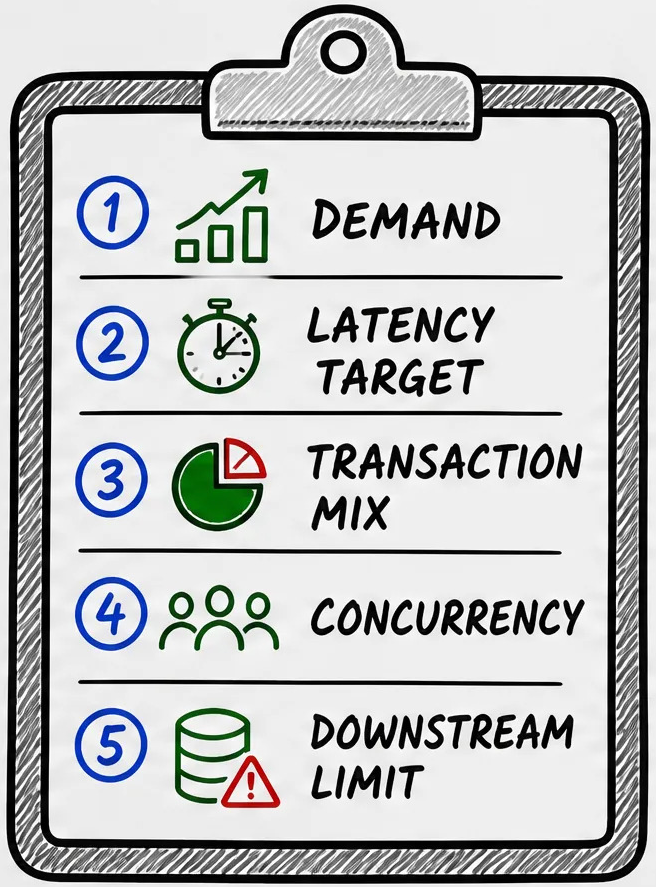




PERFORMANCE IMPROVES WHEN EVIDENCE REMOVES A CONSTRAINT.

No Universal Number:

Start with the Workload.

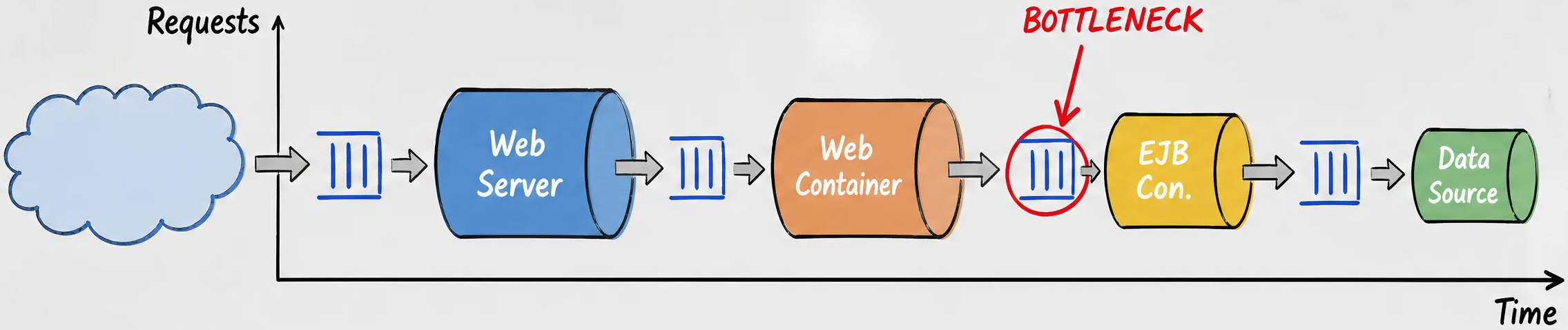


CONTEXT CHOOSES VALUES

TUNING HYPOTHESIS

WORKLOAD, NOT FOLKLORE, CHOOSES THE SETTING.

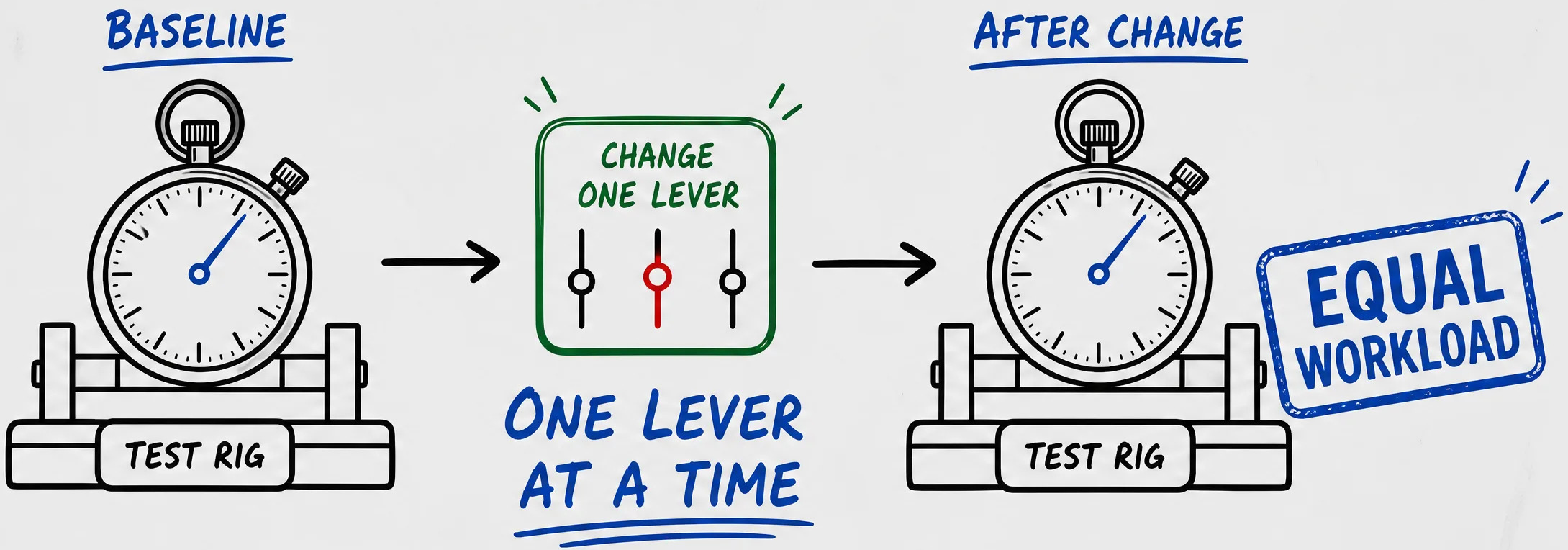
The Request Path Is a Queueing Network



Bottlenecks move downstream

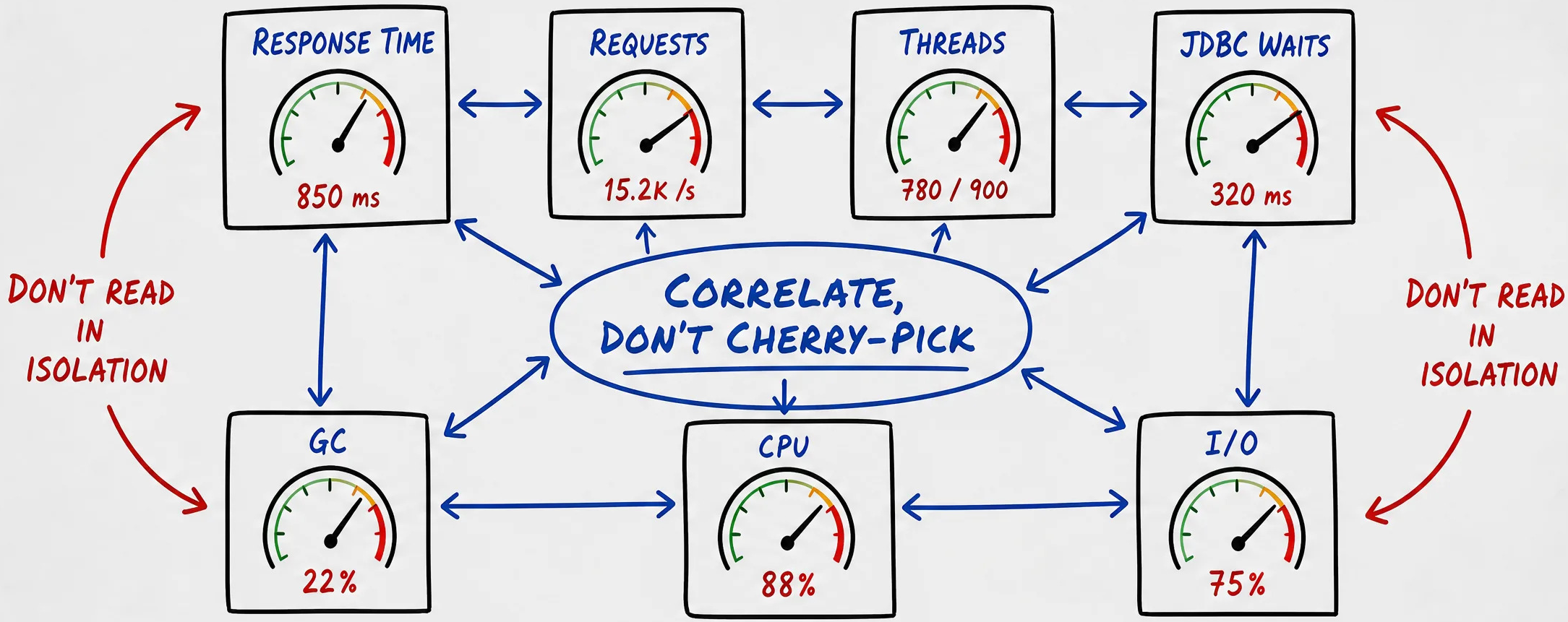
> ☆ Removing one queue exposes the next constraint.

Baseline First, Then Change One Lever.



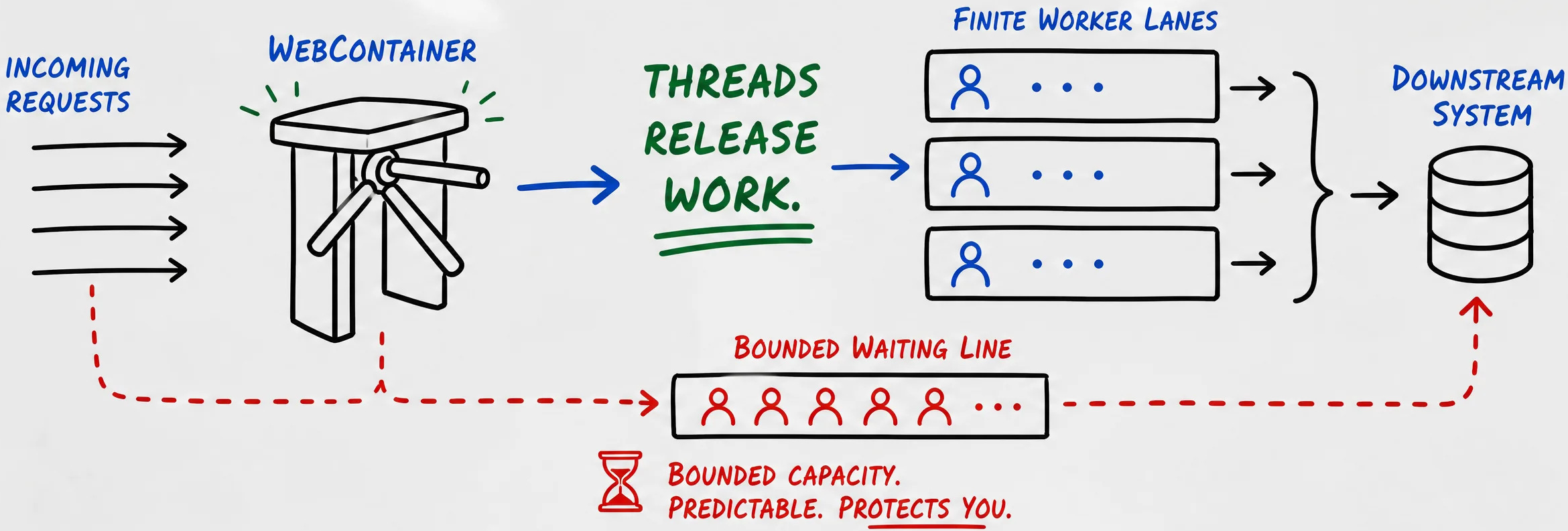
⚠ Without a comparable baseline, **faster** is only a **guess**.

Read Constraints Together.



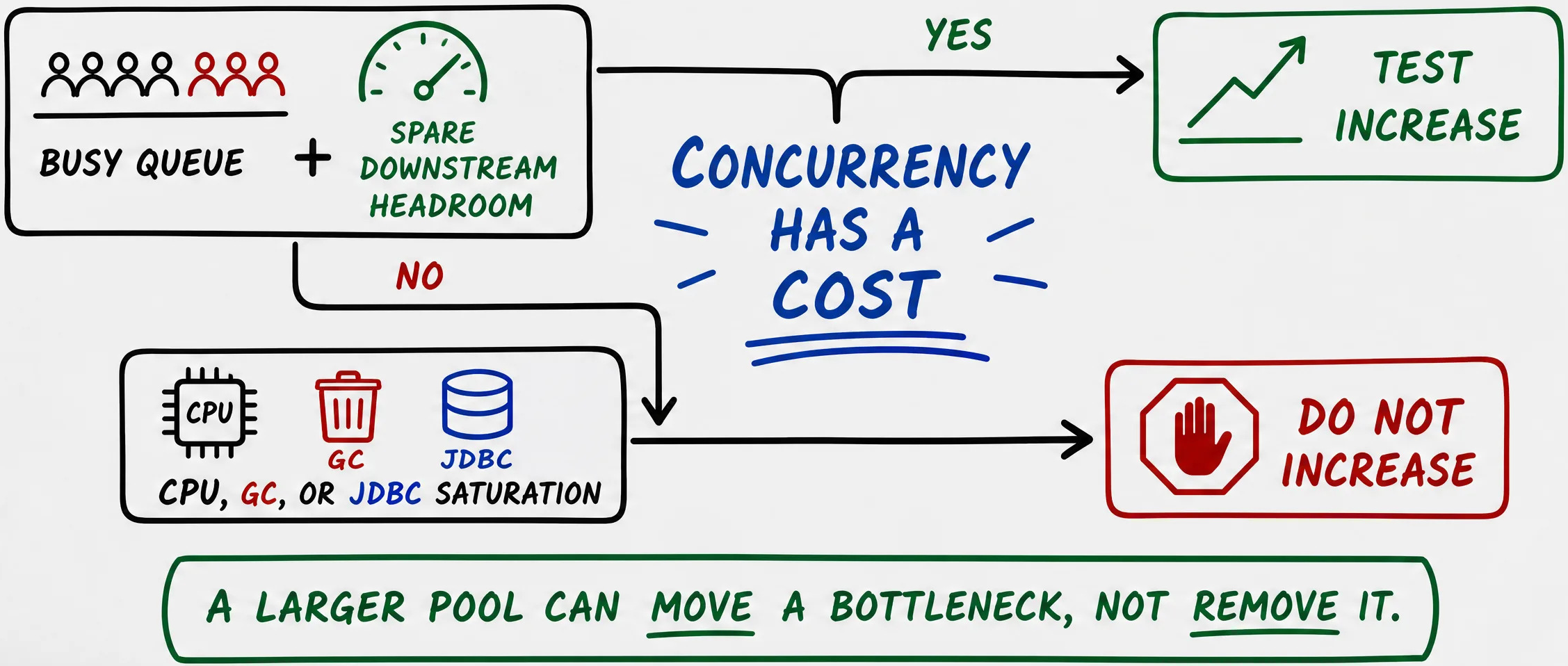
A local metric is evidence only in its system context.

WebContainer Threads are Admission Control.



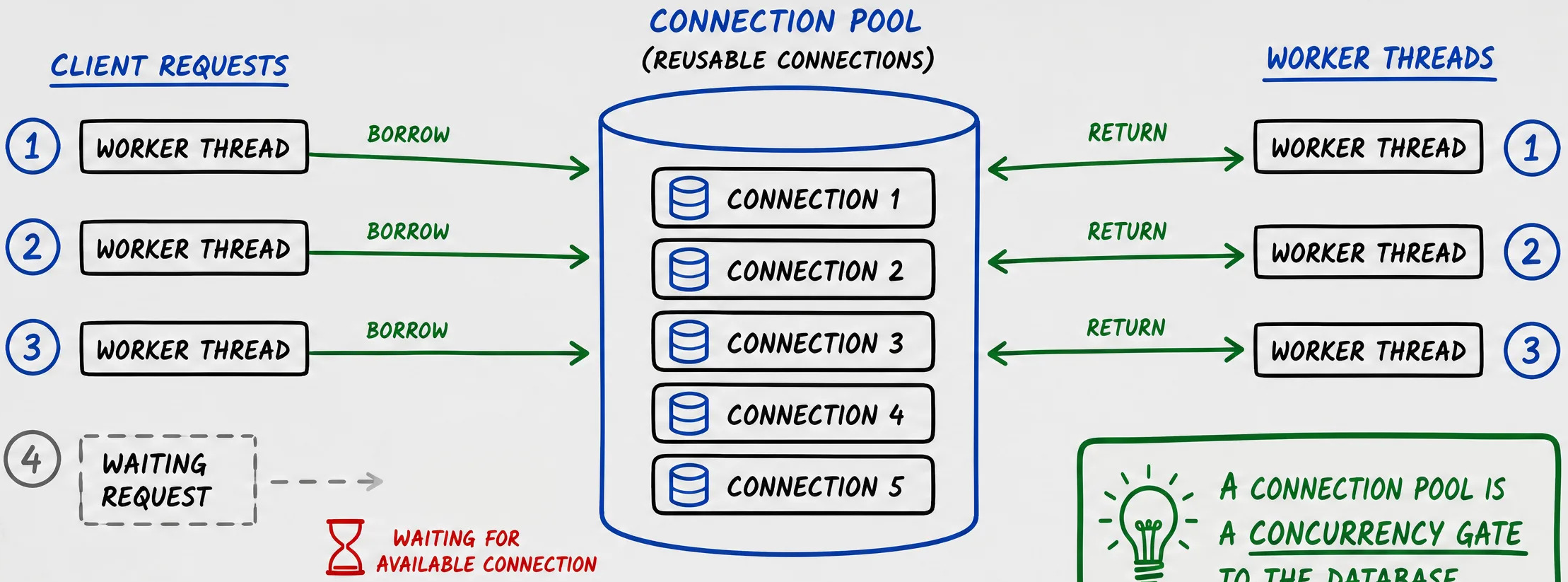
More request threads also mean more downstream pressure.

Size WebContainer Threads From Downstream Capacity.



A LARGER POOL CAN MOVE A BOTTLENECK, NOT REMOVE IT.

JDBC Pools Trade Connection Cost for Capacity.

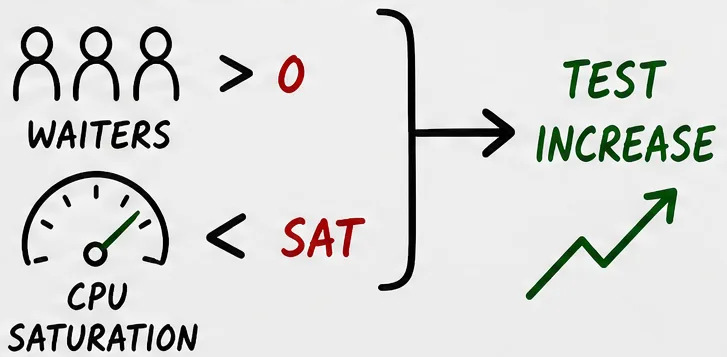


BORROW, USE, RETURN.

Size JDBC From Waiting, Use, and Backend Headroom.

Waiters tell a story.

1 Waiters above zero plus CPU below saturation



2 Low percent used



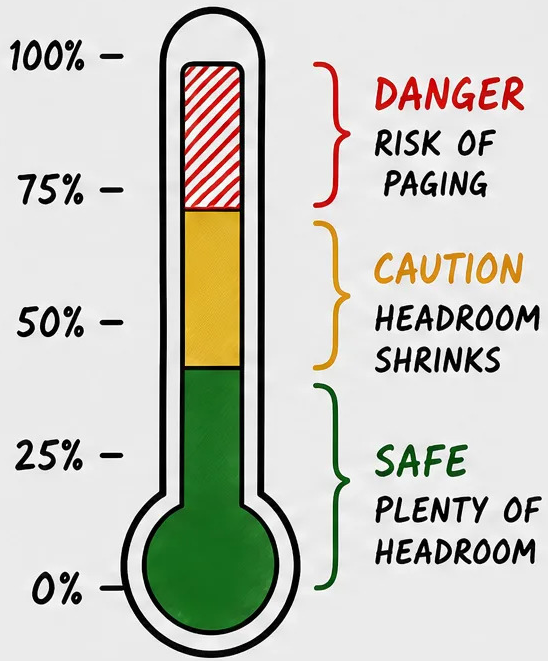
3 Multiple connections per request



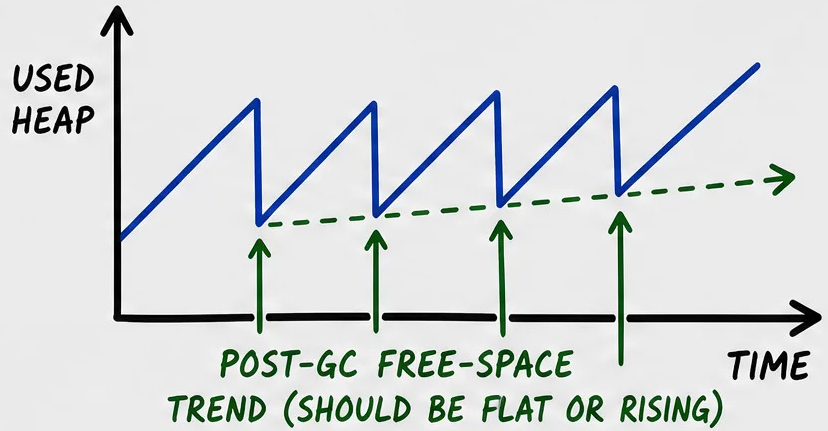
Pool size must fit <u>both</u> the request pattern and the database.

JVM Heap: Buy Throughput Without Paging.

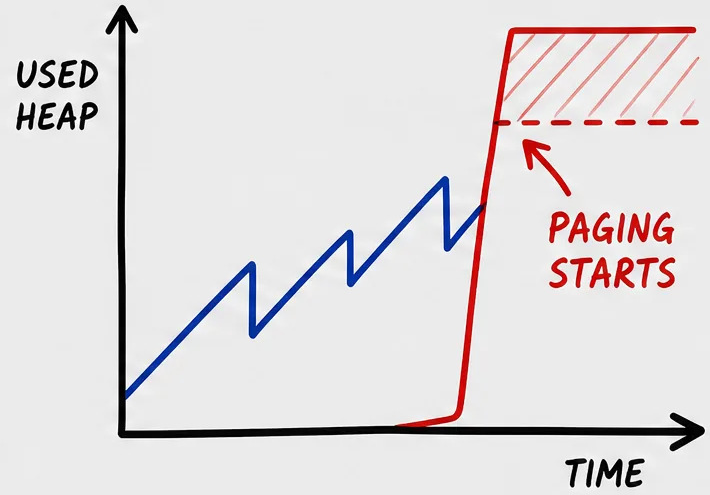
Heap Thermometer



GC Sawtooth (Normal)



Paging Cliff (Bad)





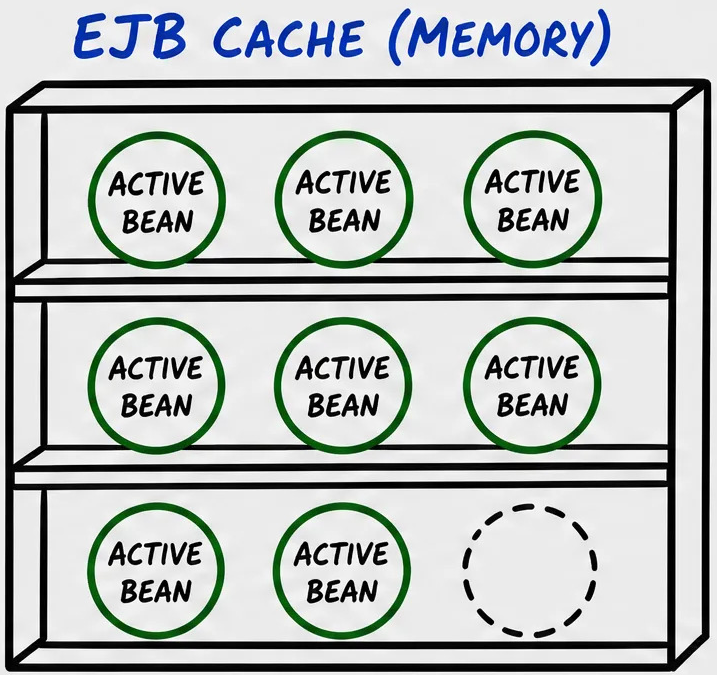
THROUGHPUT COLLAPSES

HEAP MUST STAY PHYSICAL



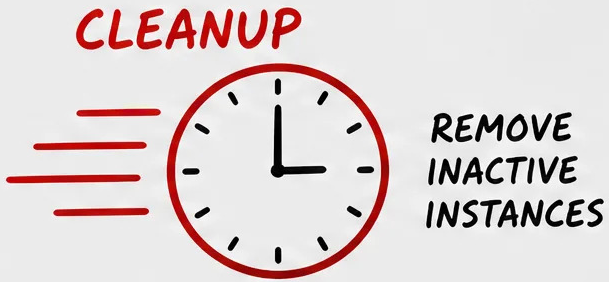
A LARGER HEAP HELPS **ONLY** WHILE MEMORY STAYS OUT OF PAGING.

EJB Cache: Memory Versus Passivation





CACHE EXPECTED ACTIVITY

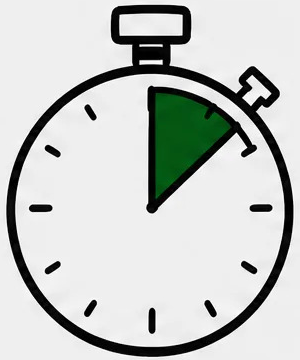


EJB CACHE SIZE IS A MEMORY DECISION BACKED BY ACTIVE-INSTANCE DEMAND.

TPV: A Short Diagnostic Window.

> TPV CONSOLE

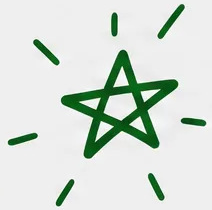
> SELECTIVE USE.



- ✓ Short duration
- ✓ Targeted intent
- ✓ High signal

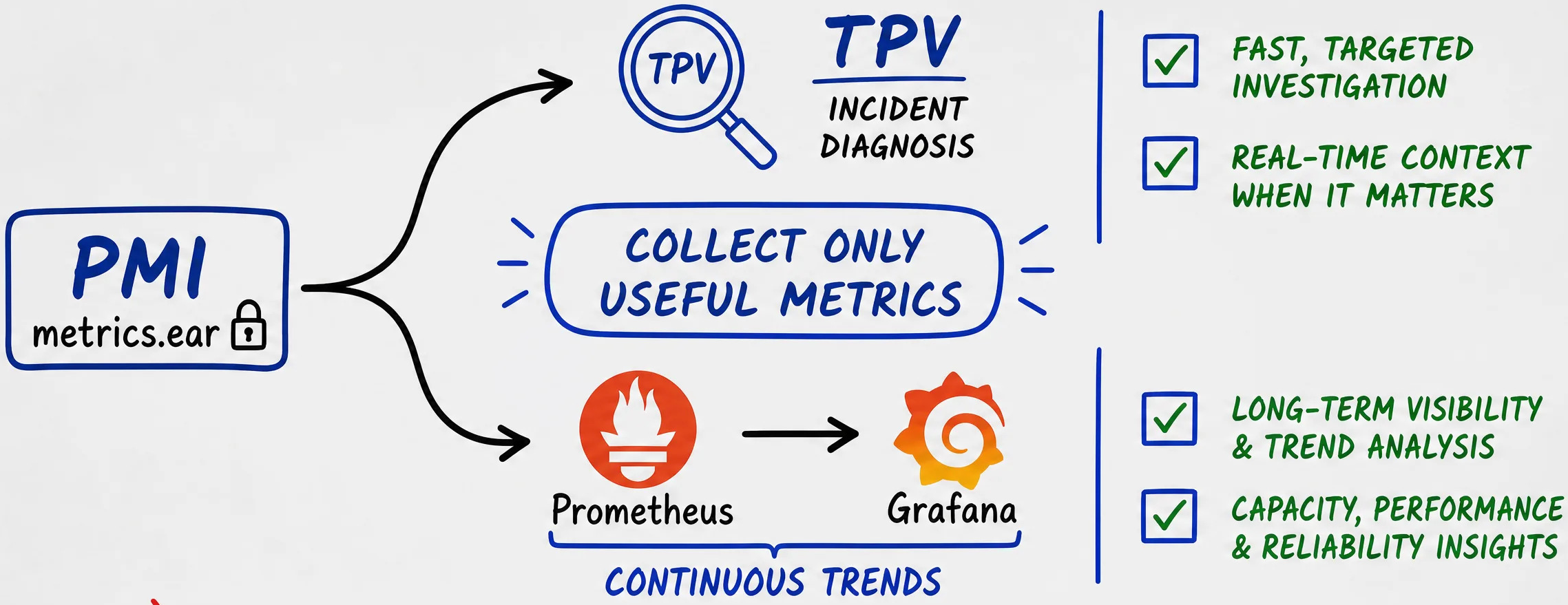
→ SAMPLE WITH INTENT →

- ADVISOR
- SUMMARY REPORTS
- PERFORMANCE MODULES
- LOG REPLAY



TPV IS FOR FOCUSED DIAGNOSIS, NOT PERMANENT SURVEILLANCE.

PMI to Prometheus: Choose the Monitoring Horizon.

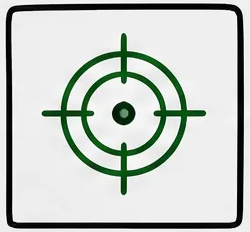


OBSERVABILITY MUST **NOT** BECOME THE NEXT BOTTLENECK.

Safe Change Procedure:

Scope, Save, Restart, Compare.

1. **Scope**



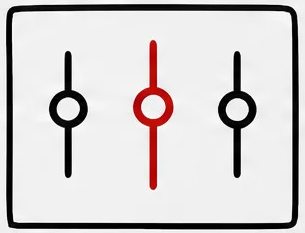
Define the goal and boundaries.

2. **Capture Baseline**



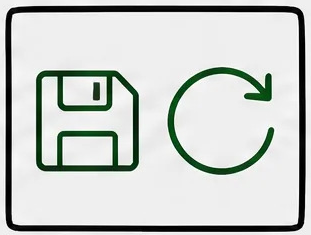
Measure and record current performance.

3. **Change One Setting**



Change only ONE setting at a time.

4. **Save / Restart as Required**



Save the change and restart as needed.

5. **Compare and Roll Back**



Compare to baseline. Roll back if no improvement.

Decision Gate



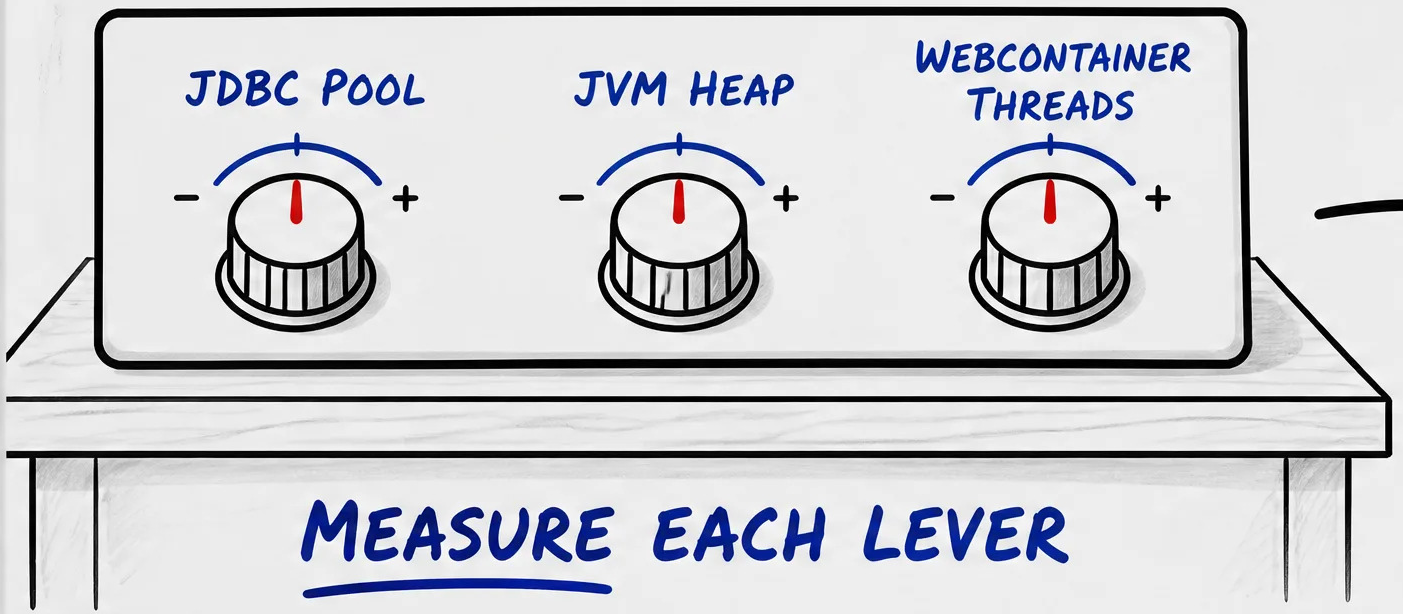
Change with an Exit



Every tuning change needs a comparison and a rollback path.

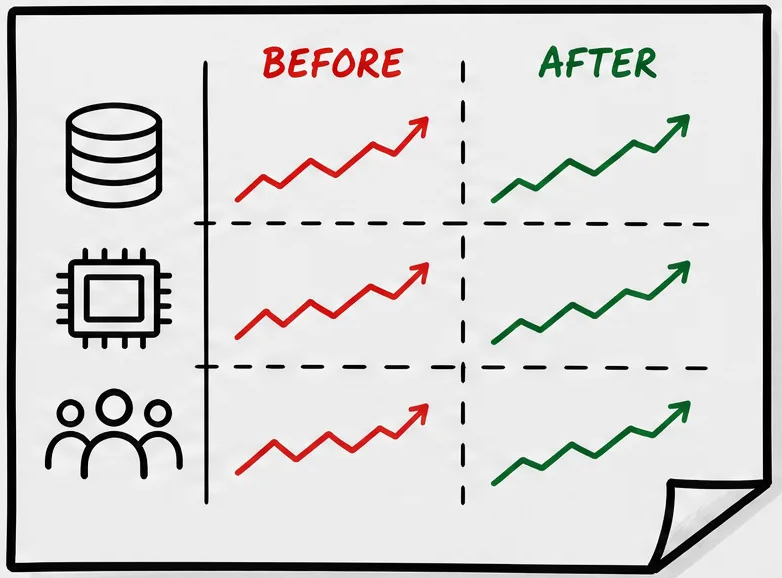
Lab: Calibrate Three Controls.

Three controls (levers)



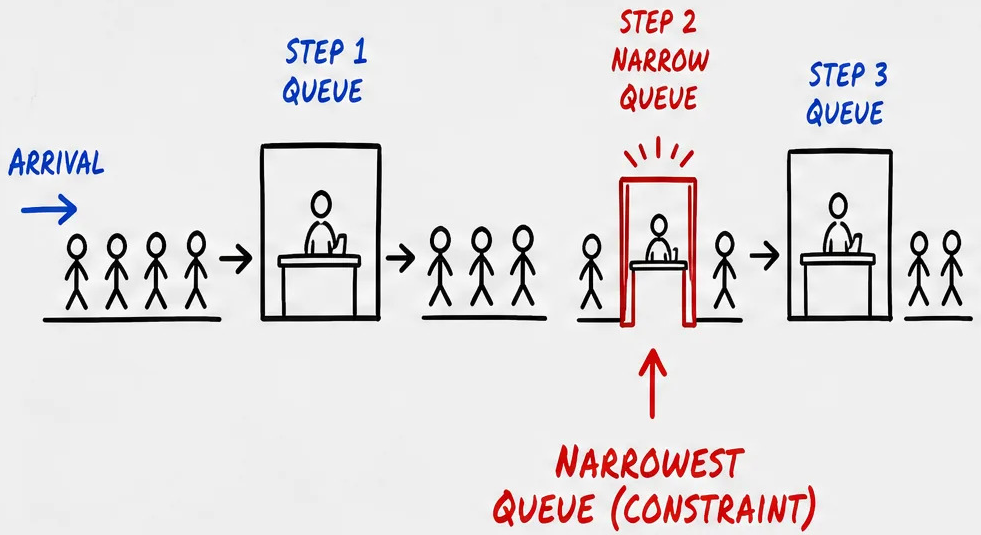
Measure each lever

Before / After Results

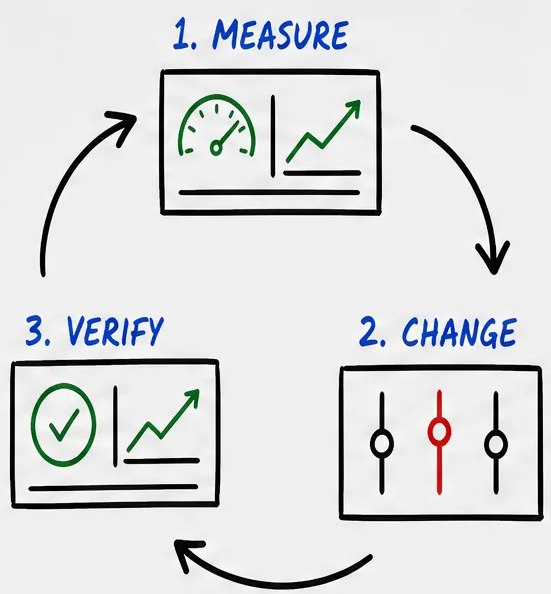


The lab result is a **defended decision**, not a **new number**.

Recap: Find the Narrowest Queue.



Tune the Constraint



Lab Checklist

- [x] Identify queues
- [x] Measure performance
- [x] Change one lever
- [x] Verify impact
- [x] Repeat

Performance improves when evidence finds the Narrowest Queue.